# CSE 311 - Data Communication

## Information Theory

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# Overview

- **Measure of Information**
- Source Coding
- Information Capacity Theorem

# Measure Of Information

- Amount of Information depends on **shock** value.

1. There will be daylight tomorrow.
2. United States invades Iran.
3. Iran invades the United States.

- Mathematically We use probability to denote shock value

# Measure Of Information

- So we denote measure of information as below:

$$I \sim \log \frac{1}{P}$$

## Average Information per Message: Entropy of a Source

Consider a memoryless source m emitting messages $m_1, m_2, \ldots, m_n$ with probabilities $P_1, P_2, \ldots, P_n$, respectively ($P_1 + P_2 + \cdots + P_n = 1$). A **memoryless source** implies that each message emitted is independent of the previous message(s). By the definition in Eq. (13.3) [or Eq. (13.4)], the information content of message $m_i$ is $I_i$, given by

$$I_i = \log \frac{1}{P_i} \quad \text{bits} \tag{13.6}$$

The probability of occurrence of $m_i$ is $P_i$. Hence, the mean, or average, information per message emitted by the source is given by $\sum_{i=1}^{n} P_i I_i$ bits. The average information per message of a source m is called its **entropy**, denoted by $H(\mathrm{m})$. Hence,

$$H(\mathrm{m}) = \sum_{i=1}^{n} P_i I_i \quad \text{bits}$$

$$= \sum_{i=1}^{n} P_i \log \frac{1}{P_i} \quad \text{bits} \tag{13.7a}$$

$$= -\sum_{i=1}^{n} P_i \log P_i \quad \text{bits} \tag{13.7b}$$

# Measure of Information

- Maximum entropy when all probabilities are equal
- See class notes for proof

# Overview

- Measure of Information
- **Source Coding**
  - Huffman Coding
  - Lempel-Ziv Source Coding
- Information Capacity Theorem

# Definition

- Code Efficiency $$\eta = \frac{H(\mathrm{m})}{L}$$

- Redundancy $$\gamma = 1 - \eta$$

# Huffman coding

| Original Source | | Reduced Sources | | | |
|---|---|---|---|---|---|
| **Messages** | **Probabilities** | $S_1$ | $S_2$ | $S_3$ | $S_4$ |
| $m_1$ | 0.30 | 0.30 | 0.30 | 0.43 | 0.57 |
| $m_2$ | 0.25 | 0.25 | 0.27 | 0.30 | 0.43 |
| $m_3$ | 0.15 | 0.18 | 0.25 | 0.27 | |
| $m_4$ | 0.12 | 0.15 | 0.18 | | |
| $m_5$ | 0.08 | 0.12 | | | |
| $m_6$ | 0.10 | | | | |

**TABLE 13.2**

| Original Source | | | Reduced Sources | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Messages** | **Probabilities** | **Code** | $S_1$ | | $S_2$ | | $S_3$ | | $S_4$ | |
| $m_1$ | 0.30 | **00** | 0.30 | **00** | 0.30 | **00** | 0.43 | **1** | 0.57 | **0** |
| $m_2$ | 0.25 | **10** | 0.25 | **10** | 0.27 | **01** | 0.30 | **00** | 0.43 | **1** |
| $m_3$ | 0.15 | **010** | 0.18 | **11** | 0.25 | **10** | 0.27 | **01** | | |
| $m_4$ | 0.12 | **011** | 0.15 | **010** | 0.18 | **11** | | | | |
| $m_5$ | 0.08 | **110** | 0.12 | **011** | | | | | | |
| $m_6$ | 0.10 | **111** | | | | | | | | |

.1 A memoryless source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1, and 0.08. Find the 4-ary (quaternary) Huffman code. Determine its average word length, the efficiency, and the redundancy.

.1 A memoryless source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1, and 0.08. Find the 4-ary (quaternary) Huffman code. Determine its average word length, the efficiency, and the redundancy.

| Messages | Probabilities | Code | Reduced Sources | |
|---|---|---|---|---|
| $m_1$ | 0.30 | **0** | 0.30 | **0** |
| $m_2$ | 0.25 | **2** | 0.30 | **1** |
| $m_3$ | 0.15 | **3** | 0.25 | **2** |
| $m_4$ | 0.12 | **10** | 0.15 | **3** |
| $m_5$ | 0.10 | **11** | | |
| $m_6$ | 0.08 | **12** | | |
| $m_7$ | 0.00 | **13** | | |

A memoryless source emits messages $m_1$ and $m_2$ with probabilities 0.8 and 0.2, respectively. Find the optimum (Huffman) binary code for this source as well as for its **second-** and **third-order extensions** (i.e., for $N = 2$ and 3). Determine the code efficiencies in each case.

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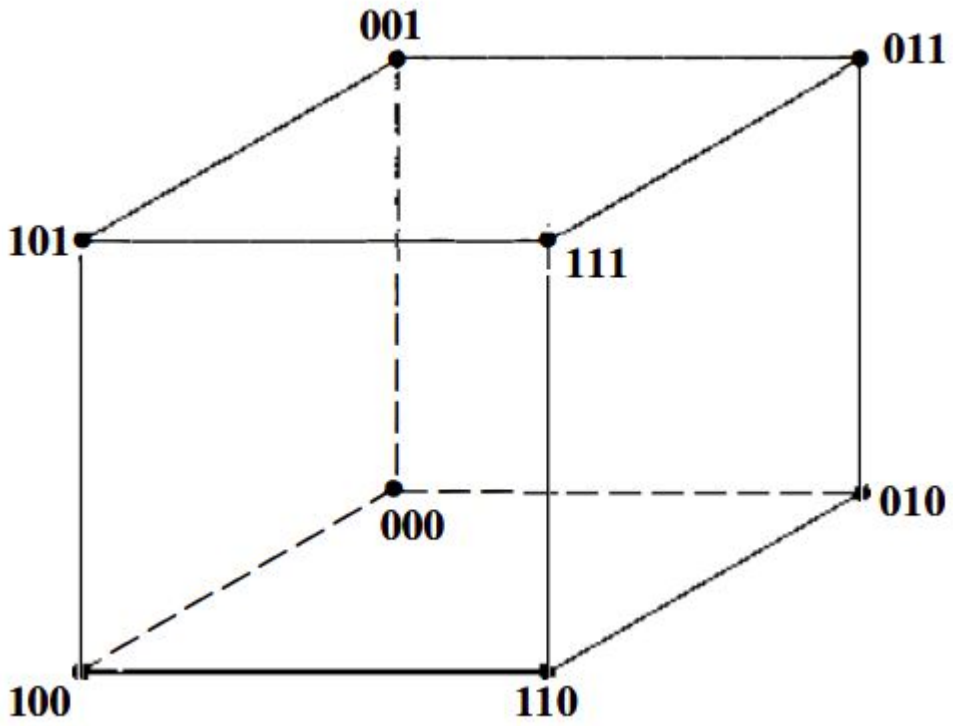
| **Messages** | **Probabilities** | **Code** | **Reduced Source** | | | |
|---|---|---|---|---|---|---|
| $m_1m_1$ | 0.64 | **0** | 0.64 | **0** | 0.64 | **0** |
| $m_1m_2$ | 0.16 | **11** | 0.20 | **10** | 0.36 | **1** |
| $m_2m_1$ | 0.16 | **100** | 0.16 | **11** | | |
| $m_2m_2$ | 0.04 | **101** | | | | |

A memoryless source emits messages $m_1$ and $m_2$ with probabilities 0.8 and 0.2, respectively. Find the optimum (Huffman) binary code for this source as well as for its **second-** and **third-order extensions** (i.e., for $N = 2$ and 3). Determine the code efficiencies in each case.

| Messages | Probabilities | Code |
|---|---|---|
| $m_1m_1m_1$ | 0.512 | 0 |
| $m_1m_1m_2$ | 0.128 | 100 |
| $m_1m_2m_1$ | 0.128 | 101 |
| $m_2m_1m_1$ | 0.128 | 110 |
| $m_1m_2m_2$ | 0.032 | 11100 |
| $m_2m_1m_2$ | 0.032 | 11101 |
| $m_2m_2m_1$ | 0.032 | 11110 |
| $m_2m_2m_2$ | 0.008 | 11111 |

# Error Free Transmission in Noisy Channel



# Conditions

$$\alpha T = \text{information digits}$$

$$\beta T = \text{total transmitted digits } (\beta > \alpha)$$

$$(\beta - \alpha)T = \text{check digits}$$

$$\frac{\alpha}{\beta} < C_s$$

# CHANNEL CAPACITY OF A DISCRETE MEMORYLESS CHANNEL

Average Uncertainty

$$H(\mathrm{x}|\mathrm{y}) = \sum_i \sum_j P(x_i, y_j) \log \frac{1}{P(x_i|y_j)} \quad \text{bits per symbol}$$

Channel Matrix

$$
\begin{array}{cc}
 & \text{Outputs} \\
\text{Inputs} & \begin{array}{c} \\ x_1 \\ x_2 \\ \vdots \\ x_r \end{array}
\begin{pmatrix}
y_1 & y_2 & \cdots & y_s \\
P(y_1|x_1) & P(y_2|x_1) & \cdots & P(y_s|x_1) \\
P(y_1|x_2) & P(y_2|x_2) & \cdots & P(y_s|x_2) \\
\vdots & \vdots & \cdots & \vdots \\
P(y_1|x_r) & P(y_2|x_r) & \cdots & P(y_s|x_r)
\end{pmatrix}
\end{array}
$$

Finding

uncertainty

$$
\begin{aligned}
P(x_i|y_j) &= \frac{P(y_j|x_i)P(x_i)}{P(y_j)} \\
&= \frac{P(y_j|x_i)P(x_i)}{\sum_i P(x_i, y_j)} \\
&= \frac{P(y_j|x_i)P(x_i)}{\sum_i P(x_i)P(y_j|x_i)}
\end{aligned}
$$

Mutual Information

$$I(\mathrm{x};\mathrm{y}) = H(\mathrm{x}) - H(\mathrm{x}|\mathrm{y}) \qquad \text{bits per symbol}$$

Channel Capacity

$$C_s = \max_{P(x_i)} I(\mathrm{x};\mathrm{y}) \quad \text{bits per symbol}$$

We have

$$I(\mathrm{x};\mathrm{y}) = \sum_i \sum_j P(x_i, y_j) \log \frac{1}{P(x_i)} - \sum_i \sum_j P(x_i, y_j) \log \frac{1}{P(x_i|y_j)}$$

$$= \sum_i \sum_j P(x_i, y_j) \log \frac{P(x_i|y_j)}{P(x_i)} \tag{13.20a}$$

$$= \sum_i \sum_j P(x_i, y_j) \log \frac{P(x_i, y_j)}{P(x_i)P(y_j)} \tag{13.20b}$$

Alternatively, by using Bayes' rule in Eq. (13.20a), we can express $I(\mathrm{x};\mathrm{y})$ as

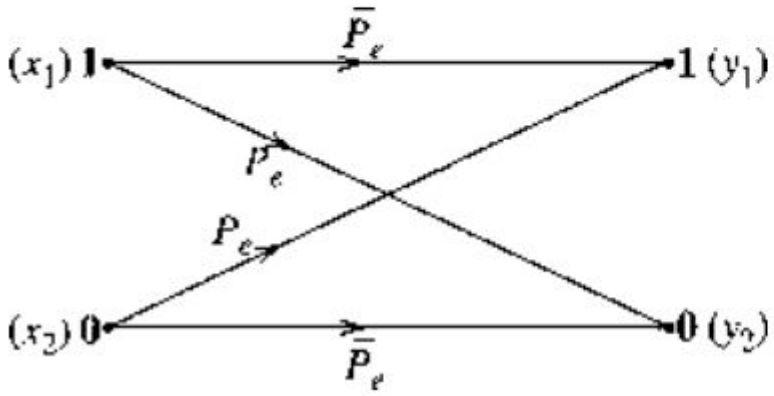
$$I(\mathrm{x};\mathrm{y}) = \sum_i \sum_j P(x_i, y_j) \log \frac{P(y_j|x_i)}{P(y_j)} \tag{13.20c}$$

or we may substitute Eq. (13.18c) into Eq. (13.20a):

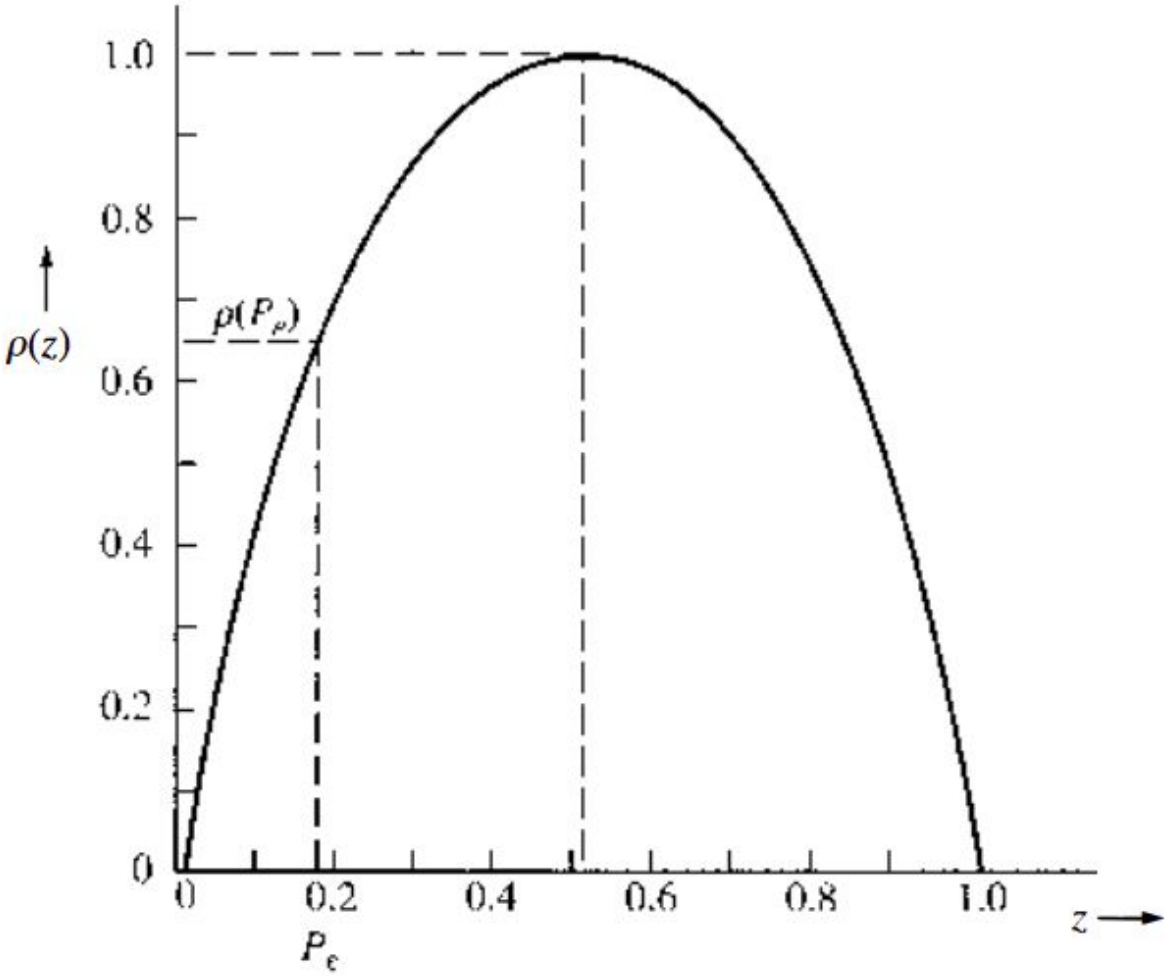
$$I(\mathrm{x};\mathrm{y}) = \sum_i \sum_j P(x_i)P(y_j|x_i) \log \frac{P(y_j|x_i)}{\sum_i P(x_i)P(y_j|x_i)} \tag{13.20d}$$

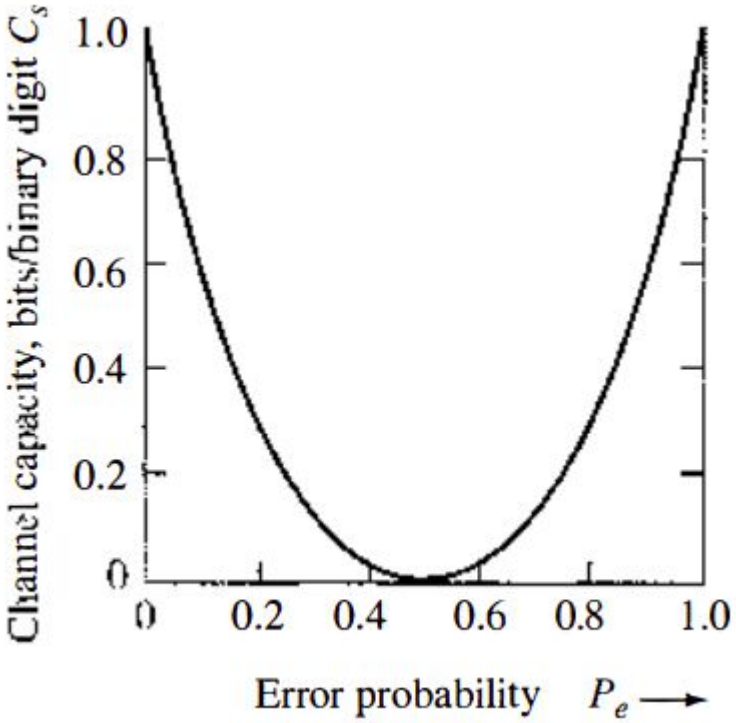
# Channel Capacity of Binary Symmetric Channel

Find the channel capacity of the BSC shown in Fig. 13.2.



$$
\begin{aligned}
I(\mathrm{x};\mathrm{y}) &= \alpha\bar{P}_e \log\left(\frac{\bar{P}_e}{\alpha\bar{P}_e + \bar{\alpha}P_e}\right) + \alpha P_e \log\left(\frac{P_e}{\alpha P_e + \bar{\alpha}\bar{P}_e}\right) \\
&\quad + \bar{\alpha}P_e \log\left(\frac{P_e}{\alpha\bar{P}_e + \bar{\alpha}P_e}\right) + \bar{\alpha}\bar{P}_e \log\left(\frac{\bar{P}_e}{\alpha P_e + \bar{\alpha}\bar{P}_e}\right) \\
&= (\alpha P_e + \bar{\alpha}\bar{P}_e) \log\left(\frac{1}{\alpha P_e + \bar{\alpha}\bar{P}_e}\right) \\
&\quad + (\alpha\bar{P}_e + \bar{\alpha}P_e) \log\left(\frac{1}{\alpha\bar{P}_e + \bar{\alpha}P_e}\right) \\
&\quad - \left(P_e \log\frac{1}{P_e} + \bar{P}_e \log\frac{1}{\bar{P}_e}\right)
\end{aligned}
$$


1.0
0.8
0.6
0.4
0.2
0
ρ(z)
ρ(P_e)
0
0.2
0.4
0.6
0.8
1.0
z
P_e


Channel capacity, bits/binary digit $C_s$
1.0
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1.0
Error probability $P_e$ ⟶

# Proof of Channel Capacity

Of all the possible sequences of $\beta T$ digits, the number of sequences that differ from a given sequence by exactly $j$ digits is $\binom{\beta T}{j}$ (see Example 8.6). Hence, $K$, the total number of sequences that differ from a given sequence by less than or equal to $\beta T P_e$ digits, is

$$K = \sum_{j=0}^{\beta T P_e} \binom{\beta T}{j} \tag{13.26}$$

$$\sum_{j=0}^{\beta T P_e} \binom{\beta T}{j} \leq 2^{\beta T \rho(P_e)} \qquad P_e < 0.5$$

Hence,

$$K \leq 2^{\beta T \rho(P_e)}$$

$$P = K2^{-(\beta-\alpha)T}$$

From Eq. (13.27) it follows that

$$P \leq 2^{-[\beta[1-\rho(P_e)]-\alpha]T}$$

Hence, as $T \to \infty$, $P \to 0$ if

$$\beta[1 - \rho(P_e)] > \alpha$$

that is, if

$$\frac{\alpha}{\beta} < 1 - \rho(P_e) \tag{13.28a}$$

But $1 - \rho(P_e)$ is $C_s$, the channel capacity of a BSC [Eq. (13.25)]. Therefore,

$$\frac{\alpha}{\beta} < C_s \tag{13.28b}$$

# CHANNEL CAPACITY OF A CONTINUOUS MEMORYLESS CHANNEL

$$
\begin{aligned}
H(\mathrm{x}) &= \lim_{\Delta x \to 0} \sum_n p(n\Delta x)\, \Delta x \log \frac{1}{p(n\Delta x)\, \Delta x} \\
&= \lim_{\Delta x \to 0} \left[ \sum_n p(n\Delta x) \Delta x \log \frac{1}{p(n\Delta x)} - \sum_n p(n\Delta x)\, \Delta x \log \Delta x \right] \\
&= \int_{-\infty}^{\infty} p(x) \log \frac{1}{p(x)}\, dx - \lim_{\Delta x \to 0} \log \Delta x \int_{-\infty}^{\infty} p(x)\, dx \\
&= \int_{-\infty}^{\infty} p(x) \log \frac{1}{p(x)}\, dx - \lim_{\Delta x \to 0} \log \Delta x
\end{aligned}
$$

A signal amplitude x is a random variable uniformly distributed in the range $(-1, 1)$. This signal is passed through an amplifier of gain 2. The output y is also a random variable, uniformly distributed in the range $(-2, 2)$. Determine the (differential) entropies $H(\mathrm{x})$ and $H(\mathrm{y})$.

We have

$$P(x) = \begin{cases} \frac{1}{2} & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P(y) = \begin{cases} \frac{1}{4} & |y| < 2 \\ 0 & \text{otherwise} \end{cases}$$

Hence,

$$H(\mathrm{x}) = \int_{-1}^{1} \frac{1}{2} \log 2 \, dx = 1 \text{ bit}$$

$$H(\mathrm{y}) = \int_{-2}^{2} \frac{1}{4} \log 4 \, dx = 2 \text{ bits}$$

# Maximum Entropy for a Given Mean Square Value of x

$$H(\mathrm{x}) = \int_{-\infty}^{\infty} p(x) \log \frac{1}{p(x)} \, dx \tag{13.33}$$

with the constraints

$$\int_{-\infty}^{\infty} p(x) \, dx = 1 \tag{13.34a}$$

$$\int_{-\infty}^{\infty} x^2 p(x) \, dx = \sigma^2 \tag{13.34b}$$

The quantities $\alpha_1, \alpha_2, \ldots, \alpha_k$ are adjustable constants, called **undetermined multipliers**, which can be found by substituting the solution of $p(x)$ [obtained from Eq. (13.37)] in Eq. (13.36). In the present case,

$$F(p, x) = p \log \frac{1}{p}$$

$$\varphi_1(x, p) = p$$

$$\varphi_2(x, p) = x^2 p$$

Hence, the solution for $p$ is given by

$$\frac{\partial}{\partial p}\left(p \log \frac{1}{p}\right) + \alpha_1 + \alpha_2 \frac{\partial}{\partial p} x^2 p = 0$$

or

$$-(1 + \log p) + \alpha_1 + \alpha_2 x^2 = 0$$

Solving for $p$, we have

$$p = e^{(\alpha_1 - 1)} e^{\alpha_2 x^2} \tag{13.38}$$

$$1 = \int_{-\infty}^{\infty} e^{\alpha_1 - 1} e^{\alpha_2 x^2} dx$$

$$= 2e^{\alpha_1 - 1} \int_0^{\infty} e^{\alpha_2 x^2} dx$$

$$= 2e^{\alpha_1 - 1} \left( \frac{1}{2} \sqrt{\frac{\pi}{-\alpha_2}} \right)$$

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 \sqrt{\frac{-\alpha_2}{\pi}} e^{\alpha_2 x^2} dx$$

$$= 2\sqrt{\frac{-\alpha_2}{\pi}} \int_0^{\infty} x^2 e^{\alpha_2 x^2} dx$$

$$= -\frac{1}{2\alpha_2}$$

$$\alpha_2 = -\frac{1}{2\sigma^2}$$

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}$$

$$e^{\alpha_1 - 1} = \sqrt{\frac{1}{2\pi\sigma^2}}$$

$$
\begin{aligned}
H(x) &= \int_{-\infty}^{\infty} p(x) \left[ \frac{1}{2} \log (2\pi\sigma^2) + \frac{x^2}{2\sigma^2} \log e \right] dx \\
&= \frac{1}{2} \log (2\pi\sigma^2) \int_{-\infty}^{\infty} p(x)\, dx + \frac{\log e}{2\sigma^2} \int_{-\infty}^{\infty} x^2 p(x)\, dx
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{2} \log (2\pi\sigma^2) + \frac{\log e}{2\sigma^2} \sigma^2 \\
&= \frac{1}{2} \log (2\pi e \sigma^2) \\
&= \frac{1}{2} \log (17.1\sigma^2)
\end{aligned}
$$

# Mutual Information

$$I(\mathrm{x};\mathrm{y}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) I(x;y)\, dx\, dy \tag{13.45a}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log \frac{p(x|y)}{p(x)}\, dx\, dy \tag{13.45b}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log \frac{1}{p(x)}\, dx\, dy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log p(x|y)\, dx\, dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x)p(y|x) \log \frac{1}{p(x)}\, dx\, dy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log p(x|y)\, dx\, dy$$

$$= \int_{-\infty}^{\infty} p(x) \log \frac{1}{p(x)} dx \int_{-\infty}^{\infty} p(y|x)\, dy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log p(x|y)\, dx\, dy$$

Note that

$$\int_{-\infty}^{\infty} p(y|x)\, dy = 1 \qquad \text{and} \qquad \int_{-\infty}^{\infty} p(x) \log \frac{1}{p(x)}\, dx = H(\mathrm{x})$$

$$H(\mathrm{x}|\mathrm{y}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log \frac{1}{p(x|y)}\, dx\, dy$$

$$I(\mathrm{x};\mathrm{y}) = H(\mathrm{x}) + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log p(x|y)\, dx\, dy \tag{13.46a}$$

$$I(\mathrm{x};\mathrm{y}) = H(\mathrm{x}) - H(\mathrm{x}|\mathrm{y})$$

$$= H(\mathrm{x}) - \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y) \log \frac{1}{p(x|y)}\, dx\, dy \tag{13.46b}$$

Substituting Eqs. (13.50) and (13.51) into Eq. (13.45b), we obtain

$$I(\mathrm{x};\mathrm{y}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x)p(y|x)\log\left(\frac{p(y|x)}{\int_{-\infty}^{\infty} p(x)p(y|x)\,dx}\right)dx\,dy \tag{13.52}$$

The conditional probability density $p(y|x)$ is characteristic of a given channel. Hence, for a given channel specified by $p(y|x)$, $I(\mathrm{x};\mathrm{y})$ is a function of the input probability density $p(x)$ alone. Thus,

$$C_s = \max_{p(x)} I(\mathrm{x};\mathrm{y})$$

If the channel allows the transmission of $K$ values per second, then $C$, the channel capacity per second, is given by

$$C = KC_s \text{ bit/s} \tag{13.53}$$

Just as in the case of discrete variables, $I(\mathrm{x};\mathrm{y})$ is symmetrical with respect to x and y for continuous random variables. This can be seen by rewriting Eq. (13.45b) as

$$I(\mathrm{x};\mathrm{y}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}\,dx\,dy \tag{13.54}$$

This equation shows that $I(\mathrm{x};\mathrm{y})$ is symmetrical with respect to x and y. Hence,

$$I(\mathrm{x};\mathrm{y}) = I(\mathrm{y};\mathrm{x})$$

From Eq. (13.48) it now follows that

$$I(\mathrm{x};\mathrm{y}) = H(\mathrm{x}) - H(\mathrm{x}|\mathrm{y}) = H(\mathrm{y}) - H(\mathrm{y}|\mathrm{x}) \tag{13.55}$$

# Capacity of a Band-Limited AWGN Channel

$$\mathrm{y}(t) = \mathrm{x}(t) + \mathrm{n}(t)$$

Because

$$y = x + n$$

for a given $x$, $y$ is equal to $n$ plus a constant ($x$). Hence, the distribution of y when x has a given value is identical to that of n except for a translation by $x$. If $p_{\mathrm{n}}(\cdot)$ represents the PDF of noise sample n, then

$$p(y|x) = p_{\mathrm{n}}(y - x) \tag{13.58}$$

$$\int_{-\infty}^{\infty} p(y|x) \log \frac{1}{p(y|x)}\, dy = \int_{-\infty}^{\infty} p_{\mathrm{n}}(y - x) \log \frac{1}{p_{\mathrm{n}}(y - x)}\, dy$$

Letting $y - x = z$, we have

$$\int_{-\infty}^{\infty} p(y|x) \log \frac{1}{p(y|x)}\, dy = \int_{-\infty}^{\infty} p_{\mathrm{n}}(z) \log \frac{1}{p_{\mathrm{n}}(z)}\, dz$$

The right-hand side is the entropy $H(\mathrm{n})$ of the noise sample n. Hence,

$$\begin{aligned} H(\mathrm{y}|\mathrm{x}) &= H(\mathrm{n}) \int_{-\infty}^{\infty} p(x)\, dx \\ &= H(\mathrm{n}) \end{aligned} \tag{13.59}$$

We have assumed that the mean square value of the signal $x(t)$ is constrained to have a value $S$, and the mean square value of the noise is $N$. We shall also assume that the signal $x(t)$ and the noise $n(t)$ are independent. In such a case, the mean square value of y will be the sum of the mean square values of x and n. Hence,

$$\overline{y^2} = S + N$$

For a given noise [given $H(n)$], $I(x; y)$ is maximum when $H(y)$ is maximum. We have seen that for a given mean square value of y ($\overline{y^2} = S + N$), $H(y)$ will be maximum if y is Gaussian, and the maximum entropy $H_{max}(y)$ is then given by

$$H_{max}(y) = \frac{1}{2} \log [2\pi e(S + N)] \tag{13.61}$$

$$\begin{aligned} I_{max}(x; y) &= H_{max}(y) - H(n) \\ &= \frac{1}{2} \log [2\pi e(S + N)] - H(n) \end{aligned}$$

For a white Gaussian noise with mean square value $N$,

$$H(n) = \frac{1}{2} \log 2\pi eN \qquad N = \mathcal{N}B$$

and

$$C_s = I_{max}(x; y) = \frac{1}{2} \log \left( \frac{S + N}{N} \right) \tag{13.62a}$$

$$= \frac{1}{2} \log \left( 1 + \frac{S}{N} \right) \tag{13.62b}$$

# Capacity of a Channel of Infinite Bandwidth

$$C = B\log\left(1+\frac{S}{N}\right)$$

$$= B\log\left(1+\frac{S}{\mathcal{N}B}\right)$$

$$\lim_{B\to\infty} C = \lim_{B\to\infty} B\log\left(1+\frac{S}{\mathcal{N}B}\right)$$

$$= \lim_{B\to\infty} \frac{S}{\mathcal{N}}\left[\frac{\mathcal{N}B}{S}\log\left(1+\frac{S}{\mathcal{N}B}\right)\right]$$

This limit can be found by noting that

$$\lim_{x\to\infty} x\log_2\left(1+\frac{1}{x}\right) = \log_2 e = 1.44$$

Hence,

$$\lim_{B\to\infty} C = 1.44\frac{S}{\mathcal{N}} \quad \text{bit/s} \tag{13.64}$$